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Changes to this document are highlighted as per this statement and are summarised below:

Section 1 - added reference to earthing philosophy drawing.
5.4, para 4 - minor clarification
6.1.3 - added reference to instrument earthing philosophy diagram
6.1.3 - aligned with 3210-8820-TN-0036 page 34
6.1.4 - clarification to align with section 5.4
6.2.1 – updated ref to superseded standard
6.2.2 – updated ref to superseded standard
6.3.1 - Note 3, amended to align with IS3043
6.4.2 - aligned with 3210-8820-TN-0036 page 35
6.5 - added clause on earthing single core cable armour.
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1.0 SCOPE

This Philosophy defines the earthing of the Electrical Installation for the Paradip refinery project and its associated facilities. This philosophy should be supplemented, as applicable, by other Standards, particularly the Electrical Earthing Philosophy Diagram, document No. 3210-8532-77-0002 and Instrument Earthing Philosophy Diagram, document No.3210-8551-63-0001.

2.0 BASIS OF DESIGN

The Earthing and Bonding System shall be designed to provide:

- Safety to personnel
- Protection against static electricity
- Protection against lightning
- To limit the voltage on a circuit when exposed to a higher voltage than that for which the circuit is designed.
- Protection against electromagnetic interference

2.1 Plant Earthing System:

The plant earthing system shall have an earthing network with required number of earth electrodes connected to it. The following shall be earthed:

- System Neutral
- Current and potential transformer secondary neutral.
- Metallic non-current carrying parts of all electrical apparatus such as transformers, Switchgears, motors, lighting/power panels, terminal boxes, control stations, lighting fixtures, receptacles etc.
- Steel structures, loading platform etc.
- Cable trays and racks, lighting mast and poles.
- Storage tanks, spheres, vessels, columns and all other process equipment.
- Electrical equipment fencing (e.g. transformer yard etc.)
- Cable shields and armour shield wire.
- Flexible earth provision for wagon truck.
- Pump handling Hydrocarbon if it's base plate is separate from motor's base plate.
- Turbo driven pump handling Hydrocarbon.

3.0 CODES AND REGULATIONS

The design and installation of the facilities and materials shall comply with the latest edition of the following standards BS 7430, BS 7671, IS 3043 and IS 1646 as tabulated in document No PDRP-8820-IN-0002 and PDRP-8820-IN-0003 as well as the requirements of Indian Electricity Rules

In case of any contradiction between various referred standard/specification/data sheet and statutory regulations, the following order of priority shall govern:

Statutory regulations

Data Sheets

Job specifications

This specification

Codes and standards

3.1 Other Standards and codes

In a situation where sufficient information/guidance is not found available in the Indian codes and standards, the British standards/IEC codes of practice latest version at the time of contract placement, shall apply. Where conflict exists between standards and codes, the more stringent shall govern.

4.0 EQUIPMENT AND MATERIALS

Equipment and materials shall be provided which conform to IEC and Indian standards.

The VENDOR shall provide all equipment which forms part of the scope, taking into consideration the external service conditions specified in the Basic Engineering Design Data document PDRP-8820-SP-0001.

5.0 SYSTEM EARTHING

5.1 220kV Incoming Supply System

The main incoming transformers and any generator transformers are wound with vector configuration YY (double star) with their neutral points resistance earthed

5.2 6.6kV Distribution System

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The 6.6kV distribution systems shall be a 3 wire system earthed via individual outdoor Neutral Earthing Resistors (NERs) local to each ~~33~~/6.9kV transformer star point. The NERs shall be selected and arranged to limit the earth fault current to the maximum earth fault current to 400A, with a short time rating of 10 seconds.

Cables shall be sized to the maximum earth fault current where appropriate.

All switchgear shall be provided with a continuous earth bar. The 6.6kV switchgear assembly shall be earthed to the substation local earth network.

The protection scheme shall be arranged to ensure that earth fault protection and discrimination is maintained under all operating conditions.

5.3 415V Distribution System

The 415V distribution system shall be a 4-wire system direct connected to earth via neutral/earth links within each MCC fed from the 6.6.kV/433V distribution transformers. The earth connection from each MCC neutral/earth link shall be connected directly to Substation earth bar via a Main Equipotential Bonding Conductor.

Cables shall be sized to the maximum earth fault current where appropriate.

All switchgear shall be provided with a continuous earth bar. The 415V switchgear assembly shall be earthed to the substation local earth network.

The protection scheme shall be arranged to ensure that earth fault protection and discrimination is maintained under all operating conditions.

5.4 Main Earth Network

Additional earth electrodes shall be installed as a minimum local to each new Main Earthing Point (NERs, Substations). The overall earth resistance shall be a maximum of 4 ohms for each electrode or electrode nest and a maximum of 1 ohm for the overall earth network. For generating stations and large substations, this value shall not be more than 1 ohm. ~~Earth electrodes shall not exceed 3m depth~~. The earth electrodes shall have clean surface, not covered by paint, enamel, grease or other materials of poor conductivity.

The Earth electrodes shall be at least of 6 m length (since there is sand backfill of 3 m at site).

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Where earth electrodes are provided purely for lightning protection then earth resistance of 5 ohms is desirable but in no case the value shall be more than 10 ohms(>10 ohm) and shall be connected to the earth network. OISD-std-180 may be referred for details on lightning protection.

Earthing of Tanks shall comply to OISD-STD-173.

Unless adequately connected to earth elsewhere (e.g. at battery limits), all utility and process pipelines should be bonded to a common conductor by means of earth bars or pipe clamps and connected to the earthing system at a point where pipelines enter or leave the hazardous area except where conflicting with the requirements of cathodic protection.

All the electrical equipment operating above 250V shall have two separate and distinct connections to earth grid. Transformers shall be provided with two dedicated earth electrodes for earthing of neutral and transformer tank earthing.

All joints in pipelines, valves, plants, storage tanks and associated facilities and equipment for petroleum shall be made electrically continuous by bonding or otherwise; the resistance value between each joint shall not exceed 1 ohm .All joints shall be protected against corrosion.

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Separate earth pits shall be provided for each of the 30m FLMs, at least two nos. per FLM, and connected separately to each FLM. All these earth pits shall be also connected to the earth grid.

Steel pipe racks in the process units and offsite area shall be earthed at every 24 metres.

An earth network shall be installed around Substations, Process Units, Structures and other Electrical Installations. The minimum size of equipotential bonds to primary electrical equipment shall be based on the short time rating of the associated switchgear. The Earth Network shall consist of a hot dip galvanised M.S flat conductor with bonds to earth rods. Each earth rod shall have a disconnection/inspection facility.

Earth Busbars shall be installed throughout the WORK SITE for the bonding of local equipment to the earth network. Each busbar shall have 2 separate connections to the earth network.

The main earth network shall be hot dip galvanised M.S flat conductor below ground and sizes for main conductors shall be indicated on the earthing layout drawing. Amount of galvanising shall be 610 gm per square meter.

5.5 Substations

In addition to the earth ring and earth electrodes external to each substation, main earth busbars shall be installed within each substation for the bonding of all equipment within the Substation, including the Main Equipotential Bonding Conductor from the neutral/earth link of all LV MCCs. The main earth busbars shall be connected directly to the local dedicated earth electrodes and to the earth network.

5.5.1 Neutral Earthing Resistors

An earth electrode arrangement shall be provided local to each NER for the Main Equipotential Bonding Conductor. The earth electrodes shall also be connected to the overall earth network. The CONTRACTOR shall size the fault rated conductor and the earth bonding conductor as appropriate.

6.0 GENERAL REQUIREMENTS

6.1 General Requirements

6.1.1 Calculations

All calculations associated with earth fault levels, cable sizing and earth network spacing shall be submitted by the CONTRACTOR to IOCL for approval.

6.1.2 Standard Installation Details

The CONTRACTOR shall prepare standard installation details for approval by IOCL.

6.1.3 Special Systems

Computer, Instrumentation and DCS Installations shall have SEPARATE earthing systems connected to the common earthing system at a single point, refer to diagram 3210-8550-63-0001.

Separate 'electronic' earthing to be provided for

i) Numerical Relays

ii) UPS

iii) DAVR

iv) Battery Chargers

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The earthing shall be insulated from the UPS Neutral terminal upto the earth pit. Hence, it should be done through insulated copper cable of appropriate cross-section as per the fault level of the UPS output source. The Earth Pit shall be zero volt/ clean earth pit and shall be of Copper plate earthing type with the earth pit depth of atleast 6m since there is 3m sand backfill at site. Two nos. of earth pits shall be prepared and paralleled for each independent UPS.

UPS earthing system should be separate from the DCS earthing system. There should be a unity ratio-isolating transformer.

No special tank isolation systems are anticipated, however if required during detailed design, the particular earthing and cathodic protection CP arrangement shall be proposed by the CONTRACTOR for IOCL approval and the CONTRACTOR shall allow for any tank isolation and CP systems in their tender.

All pipelines shall be bonded at battery limits. No cathodic protected pipelines with isolation flanges and spark gaps are anticipated, but shall be determined by the CONTRACTOR during detailed design.

6.1.4 Earthing for Safety to Personnel

All exposed conductive parts of all electrical equipment, such as motors, transformers, switchgear, motor starters, etc., shall be earthed by an equipotential bond to the earth network, or if such equipment is isolated, to one or more earth rods.

Generally two diagonally opposite bonds shall be made from each piece of major equipment to the earth network. Single bonds are acceptable for minor equipment i.e. small motors, instruments and other items with supplies less than 250V.

6.1.5 Earthing for Protection against Static Electricity

Tanks, piping and process vessels and equipment containing flammable liquids or gas shall be earthed by a connection to the earth network, or by bonding to an earthed metal structure to achieve a maximum resistance to earth of 10 ohms from any point. Bonding straps shall be applied across in line insulating devices on lines handling materials that give rise to significant amounts of static.

The subject of static accumulation caused by flow of petroleum products and the mitigation methods are dealt in detail in OISD-STD-110.and same should be referred.

6.2 Earthing for Protection against Lightning

6.2.1 Design Code

Earthing for Protection against Lightning shall comply with BS EN 62305-1 (formally BS 6651) and IS 2309

6.2.2 Zone of Protection

Any structure or equipment may be considered to be adequately shielded against lightning if it is not greater than 20m in height and it is within a cone whose apex is an earthed wire, metal structure, etc. and where the angle between the side of the cone and the vertical at the apex is not greater than 45 degrees.

Between two or more vertical conductors, spaced at a distance not exceeding twice their height, this angle may be relaxed to 60 degrees.

For structures greater than 20m in height, the rolling sphere method BS EN 62305-1 (formally BS 6651), IS 2309 shall be applied.

Calculations shall be submitted by the CONTRACTOR for IOCL approval to demonstrate the lightning protection basis of design.

6.2.3 Self-conducting Structures

Self-conducting structures such as metal framed structures, shall not be provided with air terminations or down conductors. When these are not shielded, they shall be earthed on at least two sides, by connection to the earth network, or by other suitable means in the event of a network not being available.

Steel columns and other structures on reinforced concrete foundations shall be earthed against lightning unless the foundation reinforcing and the anchor bolts are bonded to the column.

6.3 Earth Network

6.3.1 Conductors

An earth network shall be installed around Substations, Process Units, Structures, Switchracks and other Electrical Installations. The Earth Network shall consist of a main earth network (hot dip galvanised M.S flat conductor)

below ground and sizes for main conductors shall be as indicated below with bonds to earth rods.

TYPE OF EQUIPMENT	EARTH CONDUCTOR SIZE(SEE NOTE-1 & 2)
EHV &HV SUB-STATIONS	~~AS SPECIFIED~~ As Required F4
MAIN EARTH BUS/ MV & HV SWITCHGEAR INTERCONNECTIONS/POWER TRANSFORMERS	~~AS SPECIFIED~~ As Required F4
BUILDING COLUMNS	40X5 GI STRIP
STORAGE TANKS(VERTICAL& HORIZONTAL)	40X5 GI STRIP
LOADING RACKS	40X5 GI STRIP
VESSELS & HEAT EXCHANGERS	40X5 GI STRIP
PIPE RACK	40X5 GI STRIP
MOTORS ABOVE 37 KW INCLUDING HV MOTORS	16 MM(5/8") DIA. GI WIRE ROPE/40X5 MM GI STRIP
LIGHTING TRANSFORMER	16 MM(5/8") DIA. GI WIRE ROPE/40X5 MM GI STRIP
MOTORS FROM 5.5 KW TO 30 KW & WELDING RECEPTACLES	10 MM(3/8") DIA. GI WIRE ROPE
MOTORS UPTO 3.7 KW	8 SWG SOLID GI WIRE
SMALL EQUIPMENT & INSTRUMENTS	8 SWG SOLID GI WIRE
PUSH BUTTON STATIONS	8 SWG SOLID GI WIRE
LIGHTING ,POWER & INSTRUMENT PANELS	10 MM(3/8") DIA. GI WIRE ROPE
STREET LIGHT POLES	10 MM(3/8") DIA. GI WIRE ROPE
BONDING OF PIPE	25 SQ. MM INSULATED FLEXIBLE CU. CABLE.

NOTES:-

1. Earthing Conductors shall be selected and sized to BS 7430 (IS 3043).
2. Earth connections to individual equipment from nearest earth plate/grid may also be done alternatively using Copper/ Aluminium conductor PVC insulated core of size not less than half the cross section of respective power cable to equipment (motor, panel etc.) Connections shall be made using crimp type of lugs.
3. Green/yellow colour sleeving is required on protective conductors only to comply with IS3043.

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The Lighting Transformer shall be provided with two earth pits each interconnected and separately connected to the lighting transformer by insulated cables of appropriate voltage level and current rating as well as fault level.

Equipotential bonds to primary electrical equipment shall be based on the short time rating of the associated switchgear

Earth conductors shall be laid directly in the ground, preferably without breaks or joints. Underground joints, if required, shall be brazed, or welded. At convenient locations above ground main earth bars shall be provided on the earthing network to serve as points of attachment to equipotential bonds of equipment. Where possible, the earth bars shall be attached to the steel of structures.

Where the earth conductors are connected to earth electrodes, an earth inspection/disconnection pit shall be provided.

The interval between earth rods shall not exceed 25 metres, unless proven otherwise by calculation.

Where above grade joints are unavoidable, compression type joints, protected by silicone grease are acceptable.

6.4 Earth Fault Protection

6.4.1 LV Feeder Circuits

HRC Fuses (BS 88). Earth leakage protection (with contactors) where the fuse does not give adequate and selective earth fault protection.

6.4.2 LV Motor Starters

Earth fault protection, set at 5% of motor full load current or 10A, whichever is greater, shall be fitted to all motor starters.

6.4.3 Luminaires

Luminaires and small power socket outlets shall be earthed by a separate core in the supply cable.

6.5 Earthing for Protection against Electromagnetic Interference

Bonding straps shall be as short as possible. They shall have both resistance and inductance.

Bonding straps shall be resistant to corrosion, shock and vibration

Cable armour shall be bonded at both ends, wherever required. Single core cables armour shall be earthed at one end only to prevent circulating currents.

Equipment suppliers published recommendations shall be also be applied